ECCV #5522

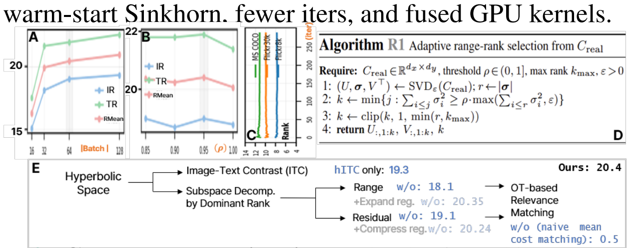
We appreciate **R1** (ky3j), **R2** (28pZ), **R3** (7WeZ) for noting ours is *well-motivated* w/ *technically-solid formulation*[R1,R2,R3], *broad analysis*[R1], and is *novel & sensible*[R2,R3].

[R1] Retrieval protocol & practical use. We respectfully apologize for the typo in L335 only: *without*→*with*. We *confirm* that all methods follow the standard VLDD retrieval protocol, using the same dot-product similarity in the shared embedding space (§4.1). Continuous synthetic text embeddings are also standard because discrete token optimization is non-differentiable (§A3.2). RAHA optimizes images in pixel space and captions as continuous embeddings, which remain compatible with the same embedding interface for contrastive finetuning. For completeness, the table below includes [69] results to be added to Table 5: [69] builds on the MTT-based [65] line by addressing modality collapse, whereas RAHA follows the distribution-matching line of [26] with hyperbolic range–residual relevance modeling in a methodologically complementary direction rather than being mutually exclusive.

# Pairs	Method	Cross-arch (Mean)	Image-δ (Mean)	Text-δ (Mean)
100/200/500	[65]	10.2 / 10.8 / 11.0	0.5 / 0.5 / 0.5	0.5 / 0.5 / 0.6
	[69]	8.1 / 8.9 / 10.7	0.6 / 0.5 / 0.5	0.5 / 0.6 / 0.5
	[26]	7.3 / 7.2 / 8.7	4.6 / 4.2 / 6.4	4.6 / 4.0 / 5.1
	Ours	6.9 / 8.7 / 12.7	4.1 / 5.0 / 6.8	3.7 / 4.5 / 6.4

[R1,R3] Retrieval gains and capacity limits. RAHA distills structured image–text relevance through a rank-aware shared subspace. It does not claim uniform dominance in every extreme-compression cell. Table 4 reports the originally published [26] scores and Sec. §A3.4/Table A4 gives the released-code same-protocol comparison: RAHA is competitive at 100 pairs and outperforms [26] at 200/500 on F30k/COCO. The 100-pair compression rate also varies by dataset: 1.7% (F8k)/0.3% (F30k)/0.8‰(COCO). Here, RAHA bring to surface an important difficulty in VLDD: dataset-scale semantics can be abstracted into decomposed range/residual bases, but a very small synthetic set may not have enough capacity to materialize those semantic modes. At extreme rate, *e.g.*, on F30k/COCO, the selected basis can be stable, but too few pairs may not populate the decomposed structure. Thus, MTT-style [65] can remain strong at the smallest budgets. As $M\uparrow$, RAHA better materializes these modes. ▷ See Table A4, Table 5, and Fig. **E**.

[R1,R3] Performance–compute trade-off. We agree RAHA is slower than [26] and will state this explicitly. At B=64, Sec.§A4/Table A9 reports 400s/iter for RAHA versus 55s/iter for [26]. A full F8k-100 run takes 1.3 vs. 0.14 GPUh, with equal 9.3GB VRAM. The overhead comes mainly from SVD on the $d\times d$ cross-cov. and Sinkhorn on the $\tilde{B}\times B$ cost matrix, not from the Lorentz lift. This is a one-time offline distillation cost, and the distilled set can be reused for downstream training. RAHA also avoids trajectory-checkpoint storage required by [63,65,69]. Thus, [26] is preferable when wall-clock efficiency is primary, while RAHA is justified when structured relevance modeling, transfer, and higher-budget scaling are prioritized. These bottlenecks are optimizable through truncated/randomized SVD, cached basis updates, low-rank or warm-start Sinkhorn. fewer iters, and fused GPU kernels.



[R2] Component attribution. Thanks for asking. Fig.**E** and A6 isolate each term under the same encoders, projections, budget, and protocol. $\mathcal{L}_{\mathrm{hITC}}$ provides base pairwise synthetic alignment. $\mathcal{L}_{\mathrm{range}}$ gives the largest single gain, showing the SVD range carries the main cross-modal relevance signal. $\mathcal{L}_{\mathrm{residual}}$ is weak alone but improves the full model when anchored by the range, acting as a controlled refinement. The geometry ablation shows the same relevance matching collapses in the all-Euclidean variant, while hyperbolic variants recover. Thus, the main gain comes from range-guided relevance transfer, with residual control and hyperbolic geometry providing stabilizing context.

[R2,R3] Rank selection & subspace stability. RAHA selects the effective rank not manually, but based on the smallest k whose cumulative squared singular-value energy of $C_{\mathrm{real}}=U\Sigma V^\top$ reaches ρ, w/ numerical-rank clipping and $k_{\max}$ (Eq.11, §A3.2, Alg.R1). This defines the range as the minimal energy-bearing image–text coupling subspace, and treats the complement as residual structure. Fig.**A** shows larger real batches improve retrieval by stabilizing the cross-covariance estimate, rather than making k simply follow the algebraic ceiling $\mathrm{rank}(C_{\mathrm{real}})\le B-1$. Fig.**B** shows ρ=0.95 is the best energy threshold, while 1.0 degrades by absorbing the low-energy residual tail. Fig.**C** further shows the selected rank remains within a dataset-specific bound over iterations, supporting stable batch-wise SVD in practice.

[R3] Image–text hierarchy, alignment & mask. RAHA does not impose an ordering between unrelated captions such as “an apple is on the desk” and “a girl is dancing.” The hierarchy is cross-modal abstraction: a generic caption such as “a dog” can describe many possible images, while a specific image or caption such as “a brown dog running through water” occupies a more detailed semantic region, as studied in hyperbolic image–text representation learning [14]. RAHA uses this as an inductive bias for VLDD, since hyperbolic geometry can keep generic concepts close to many descendants while keeping specific instances separable. The modalities are not assumed pre-aligned: NFNet and BERT provide unimodal features, and trainable projectors learn the shared retrieval space during online update (§3.1). RAHA applies hyperbolic lifting and range–residual relevance matching after this projection. Since the all-Euclidean ablation in Fig.A6 uses the same encoders and heads but collapses, the gains are not from projection pre-alignment alone. Finally, $\tilde{\mathbf{m}}_j$ only marks valid-token/padding positions, while $\tilde{\mathbf{E}}_j$ carries learnable linguistic content. **We will update as clarified in this rebuttal.**